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INDIAN STATISTICAL INSTITUTE
Mid-Semester Examination: 2025-26 (First Semester)

M. MATH. II YEAR
Commutative Algebra I

Date : 9.9.25

Maximum Marks : 40

Duration : $2\frac{1}{2}$ Hours

Throughout R denotes a commutative ring with unity.

ANSWER ANY FIVE QUESTIONS.

Each question carries 9 marks.

1. (i) Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$.
(ii) Give an example to show that the natural map $(\prod_n M_n) \otimes_R N \rightarrow \prod_n (M_n \otimes_R N)$ is not an isomorphism.
- ✓ 2. State Nagata's criterion for an integral domain to be UFD. Prove that the ring $\mathbb{C}[X, Y, Z]/(X^2 + Y^3 + Z^7)$ is a UFD.
3. Let $A = \mathbb{R}[X, Y]/(X^2 + Y^2 - 1)$. Let x and y denote the images of X and Y in A and $P = (x, y - 1)$. Prove that P is a projective A -module.
4. Prove that any finitely generated flat module over a local ring is free. Give an example to show that the finite generation hypothesis is necessary.
- ✓ 5. Prove that any finitely generated subring of $\mathbb{Q}$ is of the form $\mathbb{Z}[1/a]$ for some $a \in \mathbb{Z} \setminus \{0\}$.
- ✓ 6. Let $B = \mathbb{C}[X, Y, Z]/(XY - Z^2)$ and $A = \mathbb{C}[T^2, T^3]$. Prove that there exists a surjective $\mathbb{C}$ -algebra map ϕ from B to A . Compute $\ker \phi$.
- ✓ 7. Prove that $\mathbb{C}[X, Y]/(X^2 + Y^2 - 1)$ is a PID.
8. Prove that an integral domain R is a UFD if and only if every prime ideal of R contains a prime element. Deduce that if R is a PID, then $R[[X]]$ is a UFD using Kaplansky's result.
9. (i) Prove that any surjective R -linear map $f : M \rightarrow M$ of a finitely generated R -module M is an automorphism.
(ii) Give an example of a finitely presented R -module M and an injective R -linear map $\phi : M \rightarrow M$ which is NOT surjective.